

Karl D. Lerud

✉ karl.lerud@gmail.com

🌐 lerud <https://lerud.github.io/>

🆔 0000-0002-2156-9904

Education

- 2013 – 2019 + **Ph.D., University of Connecticut** in Psychological Sciences; Adviser: Edward W. Large
Thesis title: *Electrophysiological, neural, and perceptual aspects of pitch.*
- 2011 – 2013 + **Florida Atlantic University** in Complex Systems and Brain Sciences.
- 2009 – 2011 + **M.A., University of Wisconsin–Milwaukee** in Liberal Studies.
Thesis title: *An approach to systems-theoretic music cognition.*
- 2002 – 2006 + **B.F.A., University of Wisconsin–Milwaukee** in Music Composition and Technology.

Employment

- 2022 – present + **Postdoctoral Associate.** *Institute for Systems Research*, University of Maryland, College Park; Adviser: Jonathan Z. Simon
- 2019 – 2022 + **Postdoctoral Fellow.** *Neurology*, Baystate Health and University of Massachusetts Medical School; Adviser: Gottfried Schlaug
- 2013 – 2019 + **Research Assistant.** *Music Dynamics Laboratory*, University of Connecticut.
+ **Teaching Assistant.** *Department of Psychological Sciences*, University of Connecticut.
- 2015 – 2018 + **Scientific Programmer.** Oscilloscope, LLC.
- 2011 – 2013 + **Teaching Assistant.** *Center for Complex Systems and Brain Sciences*, Florida Atlantic University.

Publications

Journal Articles

- 1 **Lerud, K. D.**, Fisher, C., Commuri, V., Anderson, S., & Simon, J. Z. (2026a). Very Early Cortical Auditory Responses to Speech in Humans. *bioRxiv [preprint]*, 2026.06.04.730183.
<https://doi.org/10.64898/2026.06.04.730183>
- 2 Oh, Y., **Lerud, K. D.**, Hoglund, E., Klyn, N., Large, E. W., & Feth, L. L. (2023). Testing a computational model for aural detection of aircraft in ambient noise. *Journal of the Acoustical Society of America*, 154(6), 3799–3809. <https://doi.org/10.1121/10.0023933>
- 3 **Lerud, K. D.**, Hancock, R., & Skoe, E. (2023). A high-density EEG and structural MRI source analysis of the frequency following response to missing fundamental stimuli reveals subcortical and cortical activation to low and high frequency stimuli. *NeuroImage*, 279, 120330.
<https://doi.org/10.1016/j.neuroimage.2023.120330>
- 4 Ross, D. A., Shinde, A. B., **Lerud, K. D.**, & Schlaug, G. (2023). Multielectrode network stimulation (ME-NETS) demonstrated by concurrent tDCS and fMRI. *bioRxiv [preprint]*.
<https://doi.org/10.1101/2023.06.13.544867>
- 5 **Lerud, K. D.**, Vines, B. W., Shinde, A. B., & Schlaug, G. (2021). Modulating short-term auditory memory with focal transcranial direct current stimulation applied to the supramarginal gyrus. *NeuroReport*, 32(8), 702–710. <https://doi.org/10.1097/wnr.0000000000001647>

- 6 Shinde, A. B., **Lerud, K. D.**, Munsch, F., Alsop, D. C., & Schlaug, G. (2021). Effects of tDCS dose and electrode montage on regional cerebral blood flow and motor behavior. *NeuroImage*, 237. <https://doi.org/10.1016/j.neuroimage.2021.118144>
- 7 **Lerud, K. D.**, Kim, J. C., Almonte, F. V., Carney, L. H., & Large, E. W. (2019). A canonical oscillator model of cochlear dynamics. *Hearing Research*, 380, 100–107. <https://doi.org/10.1016/j.heares.2019.06.001>
- 8 **Lerud, K. D.**, Almonte, F. V., Kim, J. C., & Large, E. W. (2014). Mode-locking neurodynamics predict human auditory brainstem responses to musical intervals. *Hearing Research*, 308, 41–9. <https://doi.org/10.1016/j.heares.2013.09.010>

Conference Proceedings

- 1 **Lerud, K. D.**, Fisher, C., Commuri, V., Anderson, S., & Simon, J. Z. (2026b). Separating Cortical From Subcortical Responses to Continuous Speech Using Simultaneous EEG and MEG. In J. Holt (Ed.), *Proceedings of the Association for Research in Otolaryngology*. Oral Presentation at the Association for Research in Otolaryngology.
- 2 **Lerud, K. D.**, Fisher, C., Commuri, V., Anderson, S., Babadi, B., Kuchinsky, S. E., & Simon, J. Z. (2025). Continuous and concurrent auditory TRFs using both EEG and MEG reveal processing hierarchies during natural speech of competing speakers. In B. Cox (Ed.), *Proceedings of the Association for Research in Otolaryngology* (T131). Association for Research in Otolaryngology.
- 3 **Lerud, K. D.**, Commuri, V., Fisher, C., Anderson, S., Babadi, B., Kuchinsky, S. E., & Simon, J. Z. (2024). Continuous and concurrent auditory TRFs using both EEG and MEG reveal processing hierarchies during natural speech of competing speakers. In L. L. Colgin (Ed.), *Proceedings of the Society for Neuroscience* (460.07/F36). Society for Neuroscience.
- 4 Tonelli, L., **Lerud, K. D.**, Mechtenburg, H., Myers, E., & Skoe, E. (2023). Automated Segmentation of Brainstem, Midbrain, Thalamus, and Auditory Cortex: Test-Retest Reliability and Comparison to Manual Segmentation. In M. Kelley (Ed.), *Proceedings of the Association for Research in Otolaryngology*. Association for Research in Otolaryngology.
- 5 **Lerud, K. D.** (2019). A high-density EEG and structural MRI source analysis of the frequency following response to pitch shifted stimuli. In A. Calcus & T. Schoof (Eds.), *Frequency Following Response Workshop* (p. 15).
- 6 **Lerud, K. D.**, & Large, E. W. (2019). Source analysis of the frequency following response to pitch-shifted stimuli with high-density EEG. In P. Martens & F. Upham (Eds.), *Proceedings of the Society for Music Perception and Cognition* (p. 57).
- 7 Hoglund, E. M., Klyn, N. A., **Lerud, K. D.**, Oh, Y., Large, E. W., & Feth, L. L. (2016, October). Testing a computational model for detection of “real-world” sounds. In *Proceedings of the Acoustical Society of America* (pp. 3273–3273, Vol. 140). Acoustical Society of America. <https://doi.org/10.1121/1.4970391>
- 8 **Lerud, K. D.**, Kim, J. C., Almonte, F. V., Carney, L. H., & Large, E. W. (2015). A canonical nonlinear cochlear model. In L. J. Hood (Ed.), *Proceedings of the Association for Research in Otolaryngology* (PS–368).
- 9 **Lerud, K. D.**, Kim, J. C., & Large, E. W. (2014a). A neurodynamic account of residue pitch. In M. K. Song (Ed.), *Proceedings of the International Conference on Music Perception and Cognition* (p. 185).
- 10 **Lerud, K. D.**, Kim, J. C., & Large, E. W. (2014b). Pitch shift of the residue and its brainstem electrophysiological correlates are explained by nonlinear oscillation. In *Proceedings of the*

Acoustical Society of America (p. 2166, Vol. 135). Acoustical Society of America.

🔗 <https://doi.org/10.1121/1.4877038>

- 11 **Lerud, K. D., Kim, J. C., & Large, E. W.** (2013a). Auditory brainstem EEG, residue pitch, and nonlinear dynamical systems. In M. Schutz & F. A. Russo (Eds.), *Proceedings of the Society for Music Perception and Cognition* (2B–3.3).
- 12 **Lerud, K. D., Kim, J. C., & Large, E. W.** (2013b). Nonlinear oscillation accounts for the perception of residue pitch and its brainstem EEG correlate. In C. A. Tamminga (Ed.), *Proceedings of the Society for Neuroscience* (356.06/UU7). Society for Neuroscience.
- 13 **Lerud, K. D., Kim, J. C., & Large, E. W.** (2012). A nonlinear dynamical systems approach to pitch perception. In C. A. Tamminga (Ed.), *Proceedings of the Society for Neuroscience* (462.12/W15). Society for Neuroscience.

Teaching and training

General Psychology I Lab	+	Scientific literacy, research methods, statistics, neuroscience
Principles of Research Lab	+	Experimental design, research methods, IRB submission, data collection, regression and advanced statistical analyses, research presentation
Sensory Neuroscience Lab	+	Intro to programming in MATLAB, auditory neuroscience, fundamental signal processing
M/EEG execution and data analysis	+	Electrode, EEG net, and amplifier preparation and usage, EEG data storage and manipulation, EEG data processing and analysis, MEG marker and equipment setup, MEG digitization and preprocessing
MRI image and data analysis	+	Complete pipeline from raw MRI image volumes to analyzed data: DICOM to NIfTI conversion, preprocessing, standard space transformation, and segmentation, manual ROI creation and application, and subsequent statistical evaluations

Skills and service

Academic	+	Research methods, teaching, consultation, MATLAB training and analysis, L ^A T _E X typesetting and publishing, Git version control
Programming Languages	+	MATLAB, Python, R, L ^A T _E X, LilyPond, Git, Bash
Software	+	SPM, FSL, FreeSurfer, MRIcro/n/GL, MNE, SimNIBS, EEGLAB, FieldTrip, Brainstorm
Coding Projects	+	TRF Sensor and Source Space MEG and EEG Analysis Tools , 🔗 https://github.com/lerud/eegAndMegAnalysis
	+	GrFNN Toolbox for numerical integration of networks of nonlinear oscillators , 🔗 https://github.com/MusicDynamicsLab/GrFNNToolbox
	+	GrFNN Cochlea Toolbox for simulation of a canonical nonlinear cochlear model , 🔗 https://github.com/MusicDynamicsLab/GrFNNCochlea
Ad-hoc Reviewer	+	<i>Nonlinear Dynamics, Chaos, Journal of the Acoustical Society of America, eLife, Neuropsychologia, Brain Research, AIMS Neuroscience, Frontiers in Neuroscience, IEEE Transactions on Human-Machine Systems, eNeuro, Journal of Computational Neuroscience</i>

Awards

- 2016 + **Connecticut Institute for the Brain and Cognitive Sciences (IBaCS) Seed Grant**, \$ 22,000 for dissertation research.
- 2014 + **SEMPRE Travel Award for International Conference on Music Perception and Cognition**, \$ 662 for travel to ICMPC in Seoul, South Korea.